

	$R_{\text{LDR}} = 120 \text{ s}^{-1}$	A1
6ai	$A = \lambda N$ $460 = N \times [\ln 2 / (8.1 \times 24 \times 60 \times 60)]$ $N = 4.6 \times 10^8$	M1 A1
6aii	Number of water molecules in 1.0 kg = $(6.02 \times 10^{23}) / 0.018 = 3.3 \times 10^{25}$ Ratio = $(3.3 \times 10^{25}) / (4.6 \times 10^8) = 7.2 \times 10^{16}$	C1 A1
6aiii	$A = A_0 e^{-\lambda t}$ $170 = 460 \exp [-(\ln 2)t/8.1]$ $t = 11.6 \text{ days}$	C1 A1
6bi	If there are no other particles, β -particle would always have same energy. Range of energies means range of speeds/momenta Using conservation of energy and momentum there must be another particle to share energy/momentum	B1 B1 M1 A1
6bii	The daughter nucleus yttrium-90 also undergoes beta decay. Thus, it is difficult to deduce whether a beta particle is due to the decay of strontium-90 or yttrium-90.	B1
7ai	1. Solar energy can be affected by cloud cover. GR is only present in the day	B1